

Chapter 04 Homework Answer Sheet

Name: _____ Student ID: _____ Instructor: _____ Score: _____

Part A multiple choice (11 points)

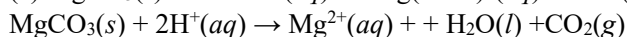
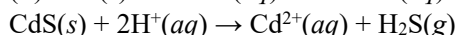
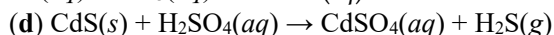
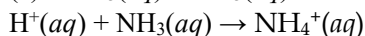
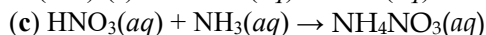
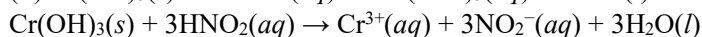
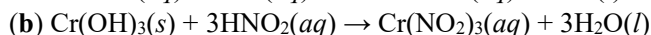
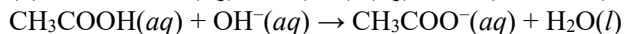
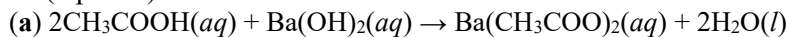
1-5: BDCCD

6-10: BBCD D/C

11: B

Part B short question (24 points)

12 (5 points)

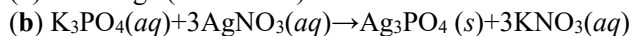


13 (4 points)

(a) +6, (b) +5, (c) +5, (d) +2

14 (4 points)

(a) Exchange (Metathesis) Reactions



(c) To determine the limiting reactant, we must examine the number

$n(\text{K}_3\text{PO}_4) = 70.5 \times 10^{-3} \text{g} \div 212.2695 \text{g/mol} = 3.32 \times 10^{-4} \text{mol}$

$n(\text{AgNO}_3) = 15.0 \times 10^{-3} \text{L} \times 0.0500 \text{mol/L} = 7.50 \times 10^{-4} \text{mol}$

Comparing the amounts of the two reactants, we find that there

$1 \text{ mol K}_3\text{PO}_4 \cong 2.26 \text{ mol AgNO}_3$

According to the part (b), $1 \text{ mol K}_3\text{PO}_4 \cong 3 \text{ mol AgNO}_3$.

Thus, there is insufficient AgNO_3 to consume the K_3PO_4 .

Therefore, AgNO_3 is limiting reactant

(d)

percentage yield $= 84.2 \div (15.0 \times 10^{-3} \times 0.0500 \div 3 \times 418.5792 \times 1000) \times 100\% = 80.5\%$

15 (2 points)

$\text{PbS}(s)$ and $\text{PbCl}_2(s)$ will precipitate.

16 (4 points)

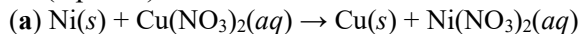
(a) acid, mixture of molecules and ions, weak electrolyte

(b) none of the above, molecules, nonelectrolyte

(c) salt, ions, strong electrolyte

(d) base, ions, strong electrolyte

17 (5 points)



(b) NR

(c) NR

